NO SPECIALIZATIONS WITHIN ETALE FIBRES OVER A POINT WITH FINITE U-FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Generalizations lift along etale morphisms. Remark remark-recall (same chapter, structural). Builds the tower of specializations through t and s.
- Points of the presentation. Properties of Spaces, lemma points-presentation (cross-chapter, named). Provides the initial point r with t(r) = u, s(r) = u'.
- Finiteness characterization. lemma-U-finite-above-x (carded L26697). The second part gives finiteness of the fibre of |U| -> |X| over x.

Source reference. Original lemma 03JS: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 485–485. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 03II: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 459–459. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03BW: focused-seeds/geometry/S9-decent-no-specializations-map-to-same-point/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $U \to X$ be an étale morphism from a scheme to X. Assume $u, u' \in |U|$ map to the same point x of $|X|$, and $u' \rightsquigarrow u$. If the pair (X, x) satisfies the equivalent conditions of Lemma 03JS then $u = u'$.*

Proof. Assume the pair (X, x) satisfies the equivalent conditions for Lemma 03JS. Let U be a scheme, $U \to X$ étale, and let $u, u' \in |U|$ map to x of $|X|$, and $u' \rightsquigarrow u$. We may and do replace U by an affine neighbourhood of u. Let $t, s : R = U \times_X U \to U$ be the étale projection maps.

Pick a point $r \in R$ with $t(r) = u$ and $s(r) = u'$. This is possible by Properties of Spaces, Lemma 03BW. Because generalizations lift along the étale morphism t

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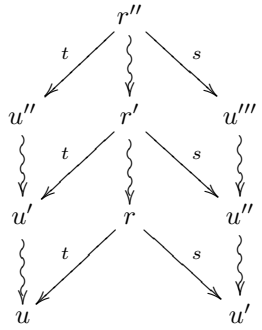
(Remark 03II) we can find a specialization $r' \rightsquigarrow r$ with $t(r') = u'$. Set $u'' = s(r')$. Then $u'' \rightsquigarrow u'$. Thus we may repeat and find $r'' \rightsquigarrow r'$ with $t(r'') = u''$. Set $u''' = s(r'')$, and so on. Here is a picture:



In Remark 03II we have seen that there are no specializations among points in the fibres of the étale morphism s. Hence if $u^{(n+1)} = u^{(n)}$ for some n, then also $r^{(n)} = r^{(n-1)}$ and hence also (by taking t) $u^{(n)} = u^{(n-1)}$. This then forces the whole tower to collapse, in particular $u = u'$. Thus we see that if $u \neq u'$, then all the specializations are strict and $\{u, u', u'', \ldots\}$ is an infinite set of points in U which map to the point x in $|X|$. As we chose U affine this contradicts the second part of Lemma 03JS, as desired. $\square$